SYS.REF.01

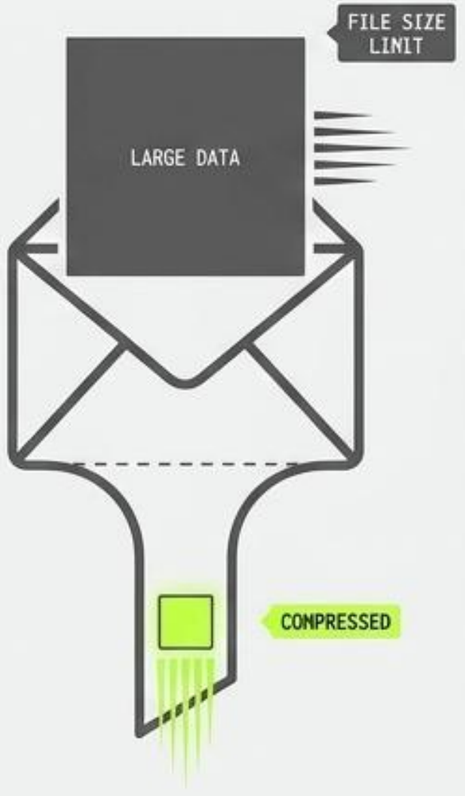
Data Compression: Mechanics & Methodologies

A technical breakdown of file size reduction, lossy vs. lossless formats, and Run Length Encoding.

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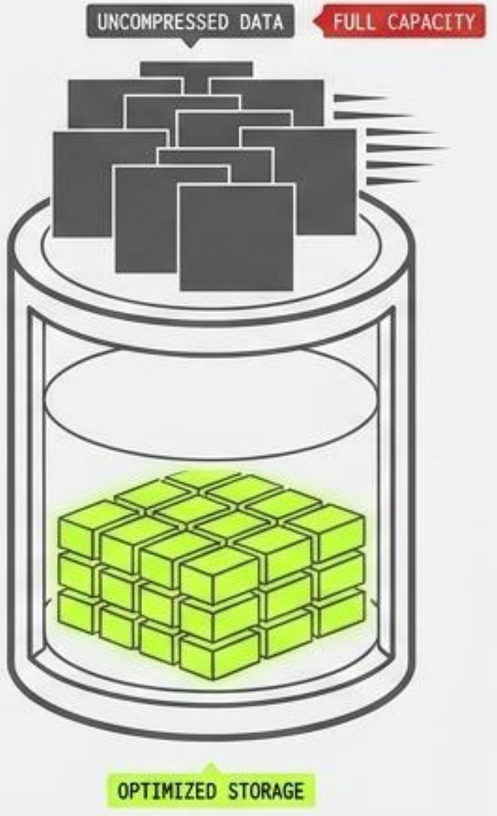
The Data Friction Problem

Email Attachment Limits



SENT: 0010110100111000 [COMPRESSED] < LIMIT

Local Storage Capacity



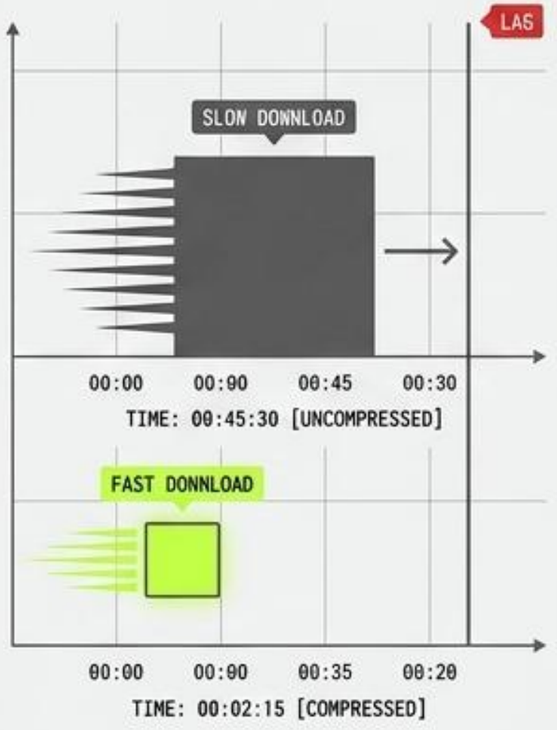
SAVED: 1.2 TB -> 400 GB [COMPRESSED] SPACE OK

USB Transfer Constraints



TRANSFER FAILED: 50 GB [TOO LARGE] USB LIMIT

Download Speeds

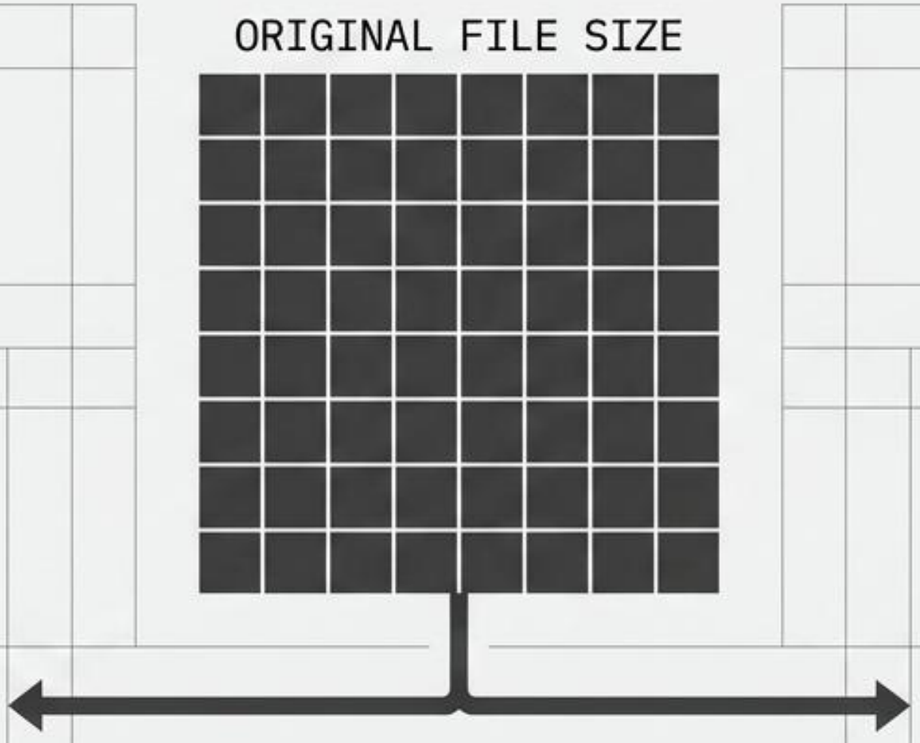


TIME: 00:02:15 [COMPRESSED]

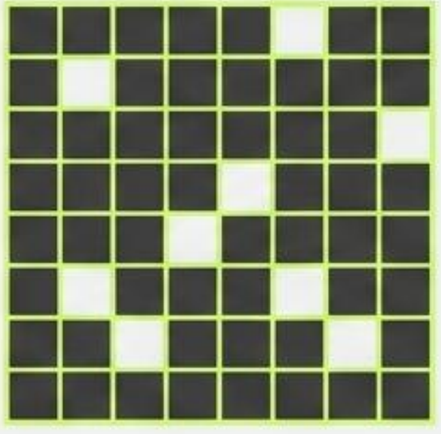
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Two Paths to Reduction

Compression mathematically alters a file's data structure so it resembles the original but requires less storage space.

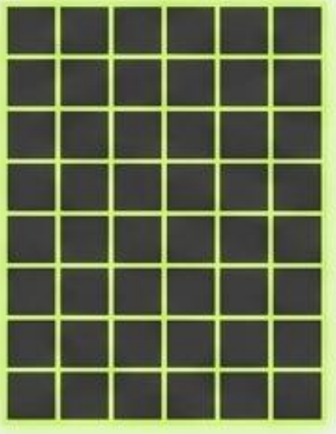


LOSSY



Delete part of the data.

LOSSLESS



Change the data structure.

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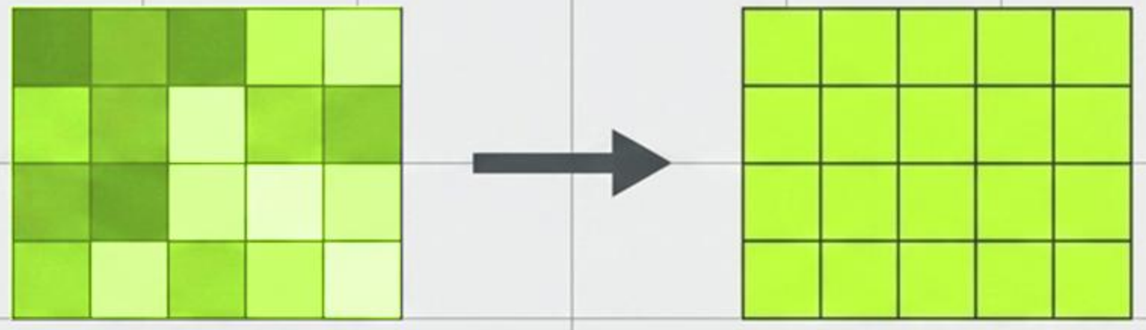
Lossy vs. Lossless Paradigms

	Lossy	Lossless
Data Deletion	Yes	No
Quality Maintained	Lower / Approximation	Identical to original
Reversibility	**Cannot restore original**	**Fully decompressible**
Target File Types	Image & Sound files	Text files, Data, Executables
Real-world Examples	JPEG, MP3	Data/Text formats

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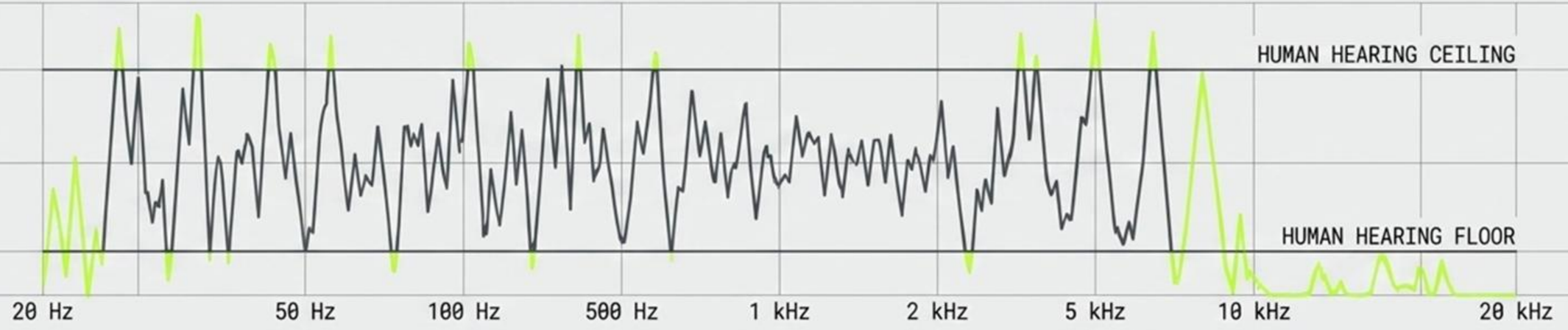
Anatomy of Lossy Media

JPEG Diagram: Pixel Duplication



Pixels with very similar colours are replaced with the same colour that can be duplicated.

MP3 Diagram: Frequency Truncation

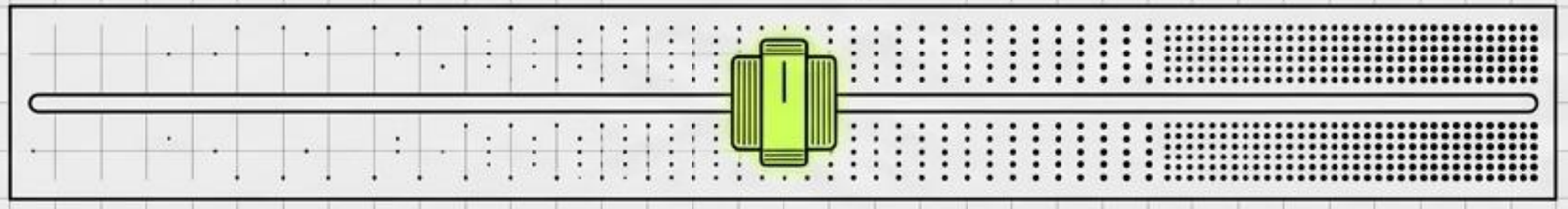


Low frequency sounds (or sounds the human ear would normally miss) are removed.

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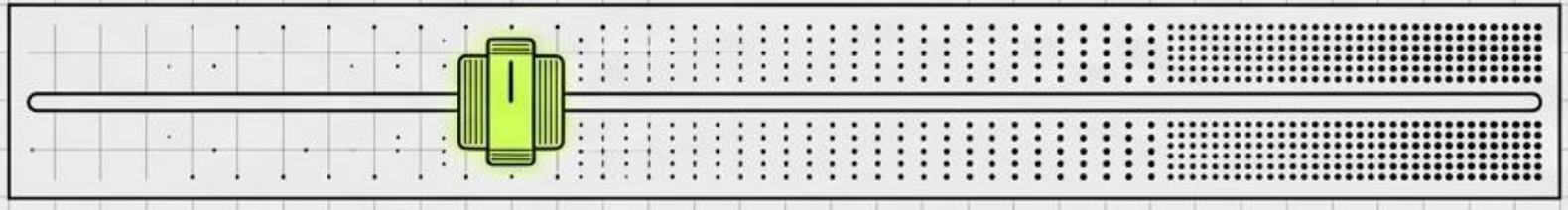
Image Settings & Data Weight

Colour Depth



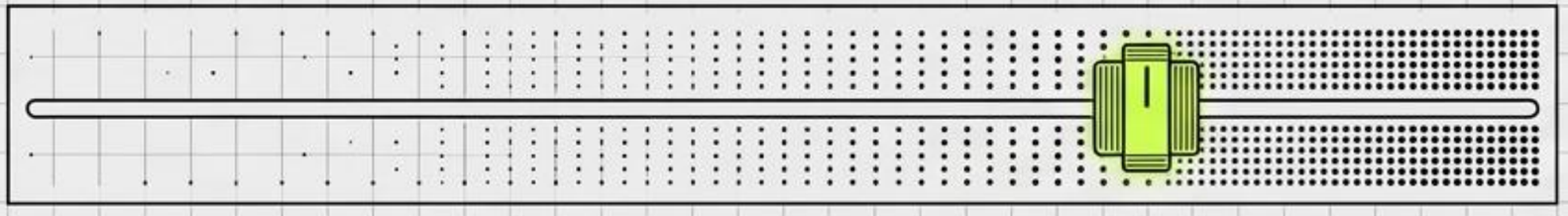
The number of colours that can be created in one pixel. Lower number of colours = less data required.

Resolution



The number of pixels in one square inch. Reducing this reduces clarity, but saves weight.

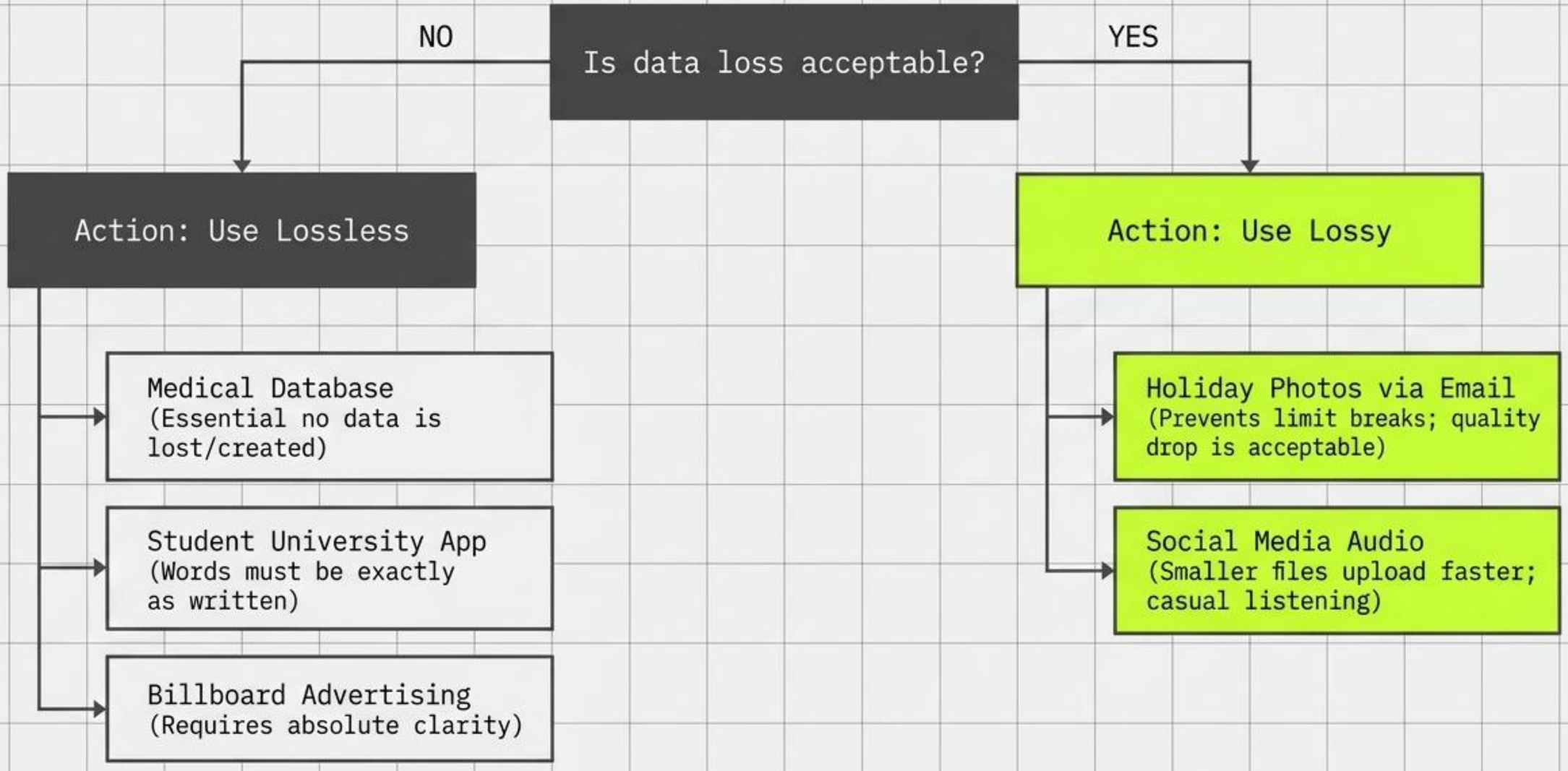
Image Size



Horizontal and vertical pixel count. Directly connected to final file size.

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The Compression Decision Tree



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Algorithmic Deep Dive: Run Length Encoding (RLE)

RLE is a lossless compression method that analyses and reduces repeated characters in a string by grouping them as byte pairs.

Original String

YYXXXXXYYYYYYYZZZZZZZ

Analysed and Grouped

RLE Version

2Y7X7Y8Z

24 bytes → 8 bytes

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Pre-Compression Analysis

Before compressing an image using RLE, the original file size must be calculated by establishing two parameters:

01

Image Dimensions

Height and Width measured in absolute pixels.

02

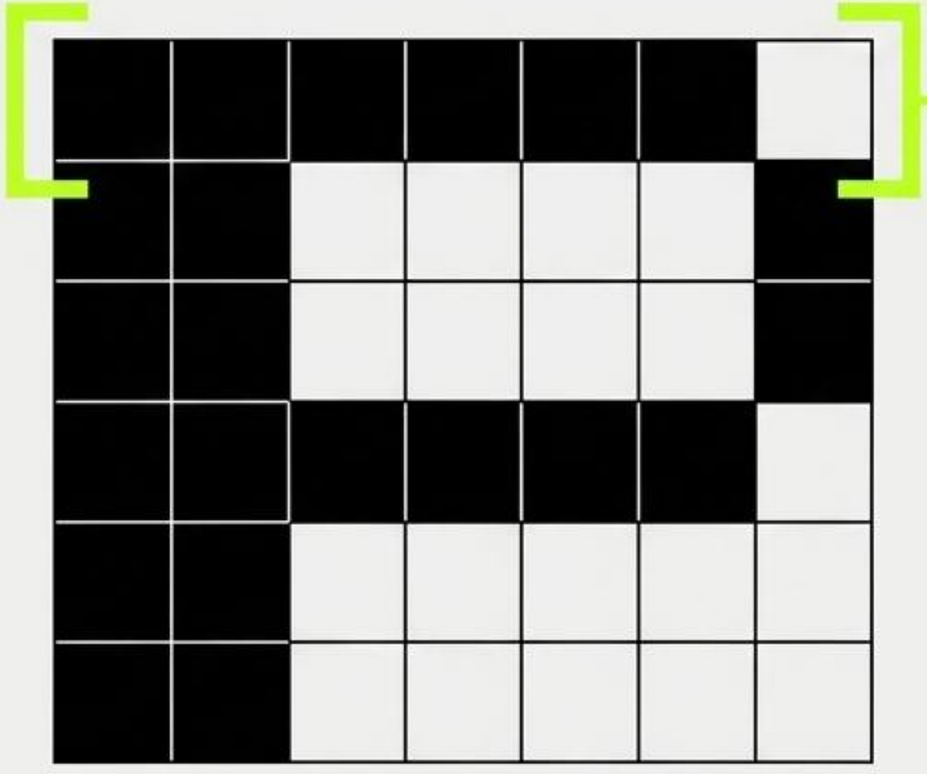
Colour Depth

Measured in bits.

Context: For the following schematics, we assume a 1-bit colour depth (black and white only).

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Compressing Simple Graphics

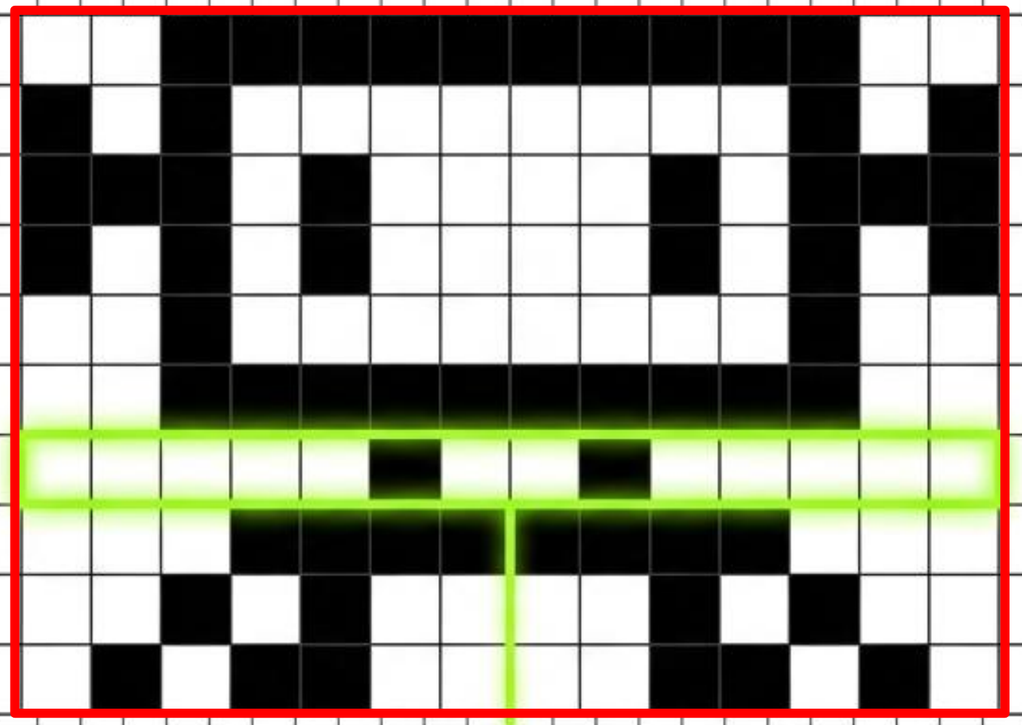


Original string	Size in bytes	RLE version	New size in bytes
bbbbbbw	7	6b1w	4

Total File Calculation: 42 bytes down to **28 bytes**.

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Compressing Complex Graphics



Row Analysis:
Original String: wwbbbbbbbbbbbbww (14 bytes)
Algorithm application: 2w10b2w (7 bytes)

Total Grid Synthesis Badge

Total grid file size reduced from 196 bytes to 182 bytes.

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The RLE Paradox



The Equation

Original String: w b w b w b (6 bytes)
RLE Encoded: 1w1b1w1b1w1b (12 bytes)

Not every image compressed via RLE results in a smaller file size. In highly alternating patterns (like the letter M), the encoding rules add more characters than they compress.

File size INCREASES from 42 to 48 bytes.

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